TECHNISCHE UNIVERSITÄT DARMSTADT

GRIS Interactive Graphics Systeme Group

MEC Lab Medical and Environmental Computing Lab

Fine-Tuning SAM2 for Temporally Consistent Semantic Segmentation of Surgical Videos with SASVi

Bachelor Thesis by Steve Jordi Nyami Gochui on November 27, 2025
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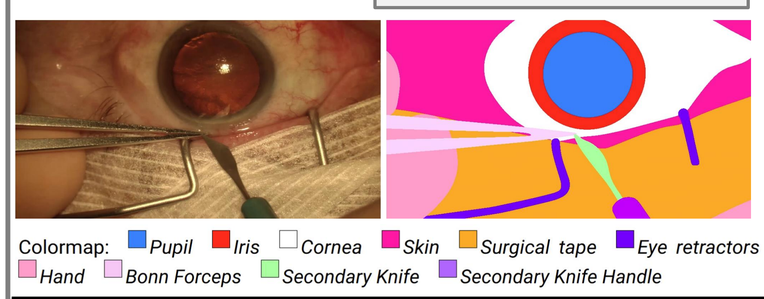
Motivation

- **Semantic segmentation** supports **tool tracking** and **surgical scene understanding**.
- Existing models like **U-Net [1]** process frames **independently**, causing **temporal flickering**.
- **SAM2 [4]** introduces memory across frames, but **its performance** on medical videos **is limited** without **domain-specific fine-tuning**.
- **Fine-tuning** enables **SAM2** to correctly track surgical structures over time, producing **temporally stable** and **accurate segmentations** across video frames.

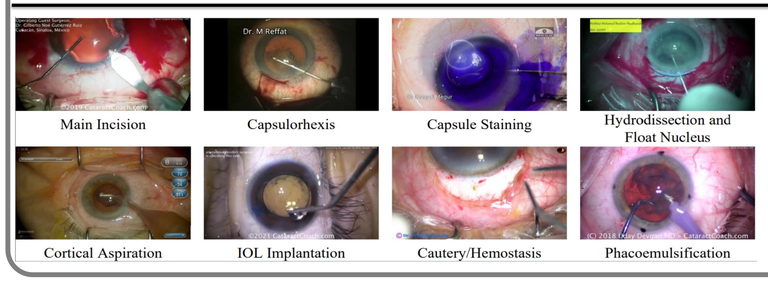
Objectives

- Fine-tune **SAM2** [4] on the **CaDIS**[2] dataset for surgical domain adaptation.
- Integrate **SAM2** into the **SASVi** [5] pipeline for **automatic overseer re-prompting**.
- Evaluate **spatial metrics (IoU, Dice)** and **SASVi specific temporal metrics**.
- Demonstrate improved **temporal consistency** over the frame-based baselines **U-Net (MONAI)** [1]
- **Test cross-domain generalization** on the **OphNet** dataset.

Datasets

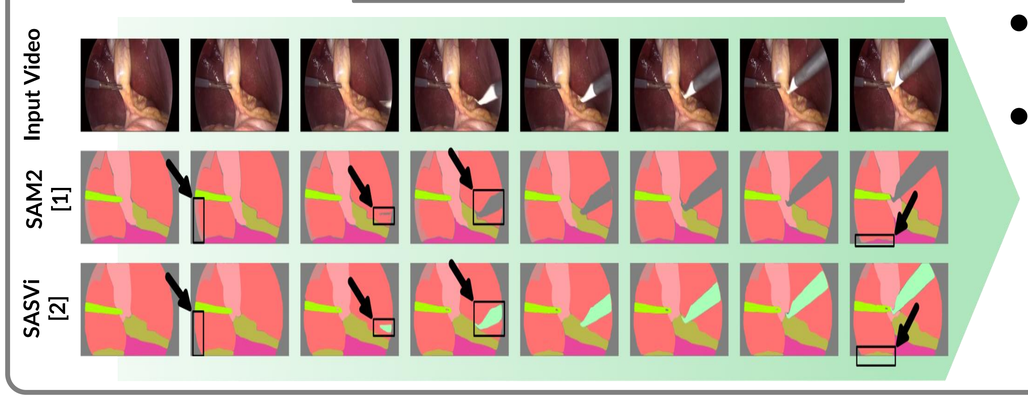


Example image frame (left) and semantic segmentation labels (right) from the Cataract dataset for Image Segmentation (CaDIS) [2]



Eight phase examples in OphNet dataset [3]

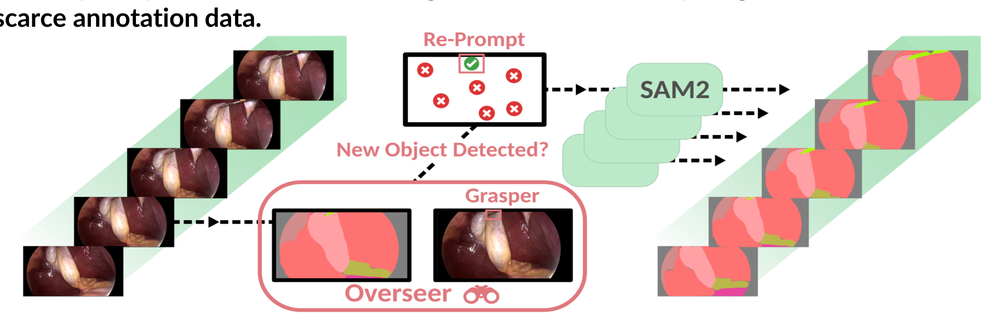
Challenges with SAM2



- Relies on manual prompting
- Fails to generalize to sections where previously tracked objects leave the scene or new objects enter.

SASVi Pipeline

Automated re-prompting mechanism based on an Overseer model for deploying SAM2 [4] for temporally consistent semantic segmentation of arbitrary surgical video domains with scarce annotation data.



Quantitative Results

Models	Test mIoU	Test Dice
SAM2.1-Tiny (Base)	0.6159	0.7451
SAM2.1-Tiny (Fine-tuned)	**0.7970**	**0.8829**
SAM2.1-Large (Base)	0.6465	0.7619
SAM2.1-Large (Fine-tuned)	**0.8209**	**0.8993**

- Base = Original SAM2.1 Weight

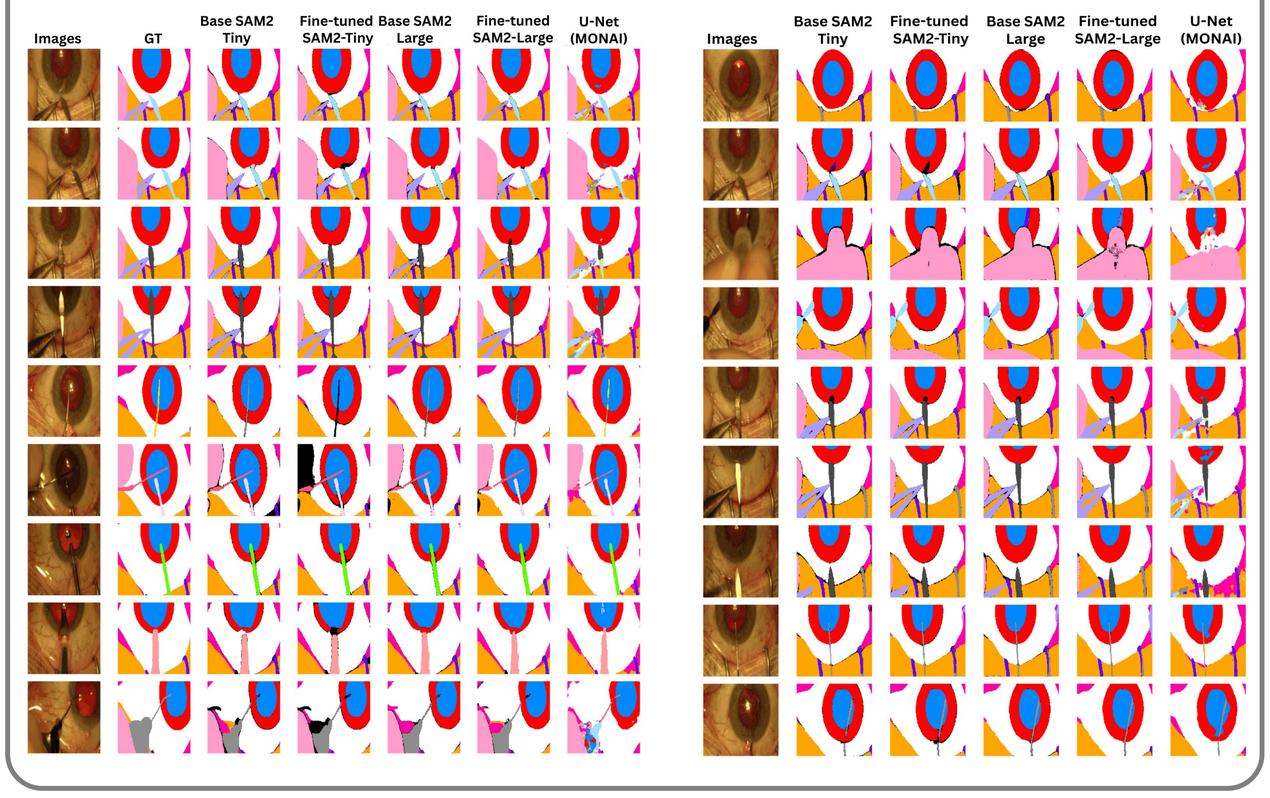
SASVi (Mask R-CNN)

Models	mIoU	Dice	Dice_{OF} (↑)	IoU_{OF} (↑)	CD_{T} (↓)	IoU_{T}
SAM2.1-Tiny (Base)	0.6872	0.7938	0.5281	0.6120	3.7153	0.6491
SAM2.1-Tiny (Fine-tuned)	0.6659	0.7778	0.5152	0.6011	3.7307	0.6367
SAM2.1-Large (Base)	0.6880	0.7929	**0.5317**	**0.6151**	**3.5952**	**0.6521**
SAM2.1-Large (Fine-tuned)	**0.6925**	**0.7958**	0.5254	0.6099	3.6887	0.6461
U-Net (Monai)	0.6736	0.7848	-	-	-	-

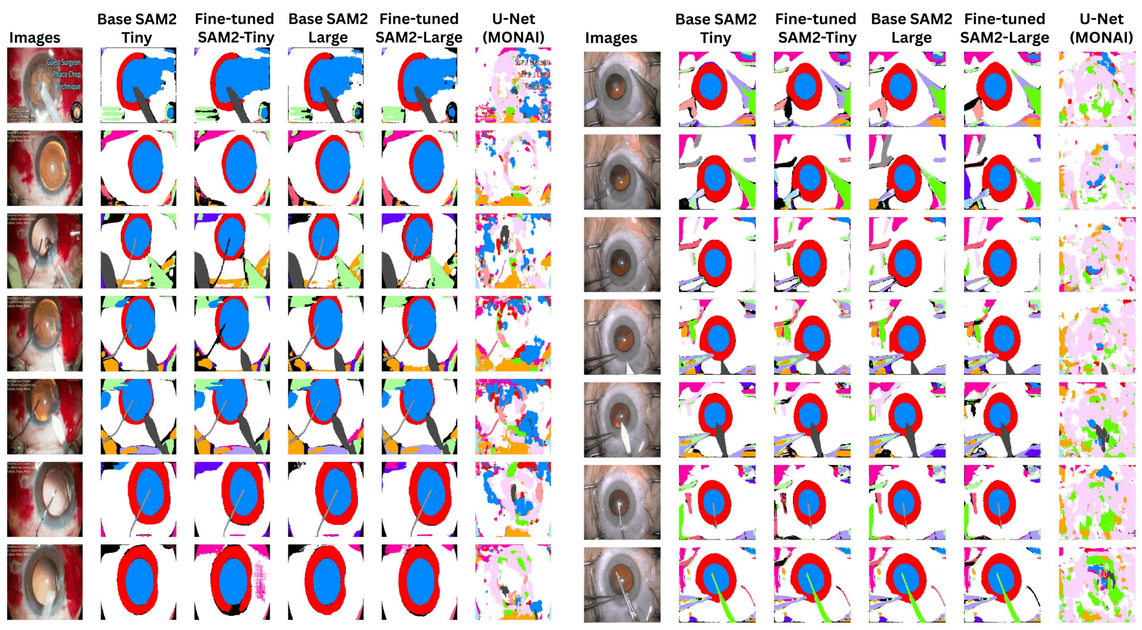
SASVi (Mask2Former)

Models	mIoU	Dice	Dice_{OF} (↑)	IoU_{OF} (↑)	CD_{T} (↓)	IoU_{T}
SAM2.1-Tiny (Base)	**0.7141**	**0.8236**	0.5398	0.6254	3.4822	0.6659
SAM2.1-Tiny (Fine-tuned)	0.6832	0.7998	0.5288	0.6151	3.4995	0.6539
SAM2.1-Large (Base)	0.7136	0.8231	**0.5436**	**0.6296**	**3.2331**	**0.6702**
SAM2.1-Large (Fine-tuned)	0.7116	0.8204	0.5365	0.6211	3.4255	0.6621
U-Net (Monai)	0.6736	0.7848	-	-	-	-

Qualitative Results



Generalization on OphNet



Conclusion

- **Fine-tuning SAM2** [4] substantially improves **spatial accuracy (IoU, Dice)** across all **17 evaluated classes.**
- Integration into the **SASVi [5]** pipeline enables **stable, temporally consistent semantic segmentation** across surgical video frames.
- The **fine-tuned SAM2** models outperform **U-Net [1]** and the **SAM2 original model weights**, especially using **SASVi** with **Mask R-CNN** as overseer.
- The approach **generalizes well to OphNet [3]** , demonstrating robustness across different surgical domains.
- The combination of **Overseer prompts + SAM2** memory provides a reliable foundation for **real-time, clinically relevant** surgical scene understanding.

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SCAN ME

[1] Olaf Ronneberger et al. “U-net: Convolutional net works for biomedical image segmentation”, 2015
[2] Maria Grammatikopoulou et al. “CaDIS: Cataract dataset for surgical RGB-image segmentation”, 2021.
[3] Ming Hu et al. “OphNet: A large-scale video benchmark for ophthalmic surgical workflow understanding”, 2024.
[4] Nikhila Ravi et al. “SAM 2: Segment Anything in Images and Videos”, 2024.
[5] Ssharvien Sivakumar et al. “SASVi: segment any surgical video”, 2025.